

Unit 406 Field Data Review

Mohawk Valley West Slope demonstration. **Cable** unit, 35.1 ac. 17 plots, BAF 20, variable-radius cruise on a 300 ft grid, 205 trees tallied. Simulated data with planted recording errors; see docs/methods.md. Plots tally all trees, so this is a pre-marking stand exam; the sale-cruise standards are applied as if the whole tallied volume were designated for cutting.

TABLE 1. STAND SUMMARY, BAF 20 CRUISE, PER ACRE

STAND DENSITY		STAND INFO	
Basal area	241 sq ft/ac (1 SE of mean, BA 2.3 %)	Type	Sierran mixed conifer
Trees per acre	71	CWHR size / density	5 / D (cover 93 %)
QMD	24.9 in	SDI	297 (50 % of max 598)
SAWTIMBER (>= 10 in)		BIOMASS (< 10 in)	
BA QMD	241 sq ft/ac 24.9 in	BA QMD	0 sq ft/ac 0.0 in
TPA	71	TPA	0
Volume	42.0 MBF/ac (8366 cu ft)	Volume	0.0 green tons/ac (0 cu ft)
TREATMENT TARGET		CRUISE DESIGN	
Leave	35 % of max SDI, about 170 sq ft/ac BA	Vol. sampling error	9.5 % at 95 % (CV 18 %, t = 2.12, n = 17)
Species (BA)	WF 82, PP 60, IC 42, DF 34, SP 22	Stratum standard	30 % of volume (R5 2409.12-2021-4, sec. 41.1): MEETS ; plots for standard 2
		Sale volume share	1.9 % of sale volume: standard shown for information (under the R5 10 % threshold)
		BA sampling error	4.9 % at 95 % (CV 9 %); reference, not the standard
		Measured trees	PP 51, WF 70, DF 29, SP 19, IC 36; R5 20 measured trees per major species (sec. 41.3; major by sale value: PP, WF, DF, IC): ok
		Plot count	17; R5 20-plot minimum for an area-based cruise (sec. 41.3): SHORT by 3

Cubic volume is total-stem CVTS by species (PNW-FIA California equations, MacLean and Berger 1976, CARB 2011 compendium), net of defect. Scribner board feet at 5.02 bf/cu ft (Keegan et al. 2010, table 2, California 2000-2006). Green tons at PP 45, WF 47, DF 38, SP 49, IC 45 lb/cu ft of wood (Miles and Smith 2009, NRS-38 table 1A). SDI in the summation form; maximum is the basal-area-weighted FVS Western Sierra species maximum (table 3.5.1, rev. 2025-09-23). The Keegan ratio is Scribner log scale per cubic foot of delivered sawlog fiber; applied to total-stem CVTS it overstates sawlog volume. Merchantable cubic volume is the consistent pairing. Sampling error at 95 % confidence, Student's t, on plot net cubic volume per acre, the quantity the standards test. Standards: Region 5 Supplement 2409.12-2021-4 to FSH 2409.12 ch. 40 (effective 2021-08-11), sec. 41.1, stratum standard 30 % of volume for tree-measurement sales (40 % scaled; the national text 41.1(5)(b) reads 40 %), binding for any stratum over 10 % of total sale volume and shown for information otherwise; sec. 41.3, 20 measured trees per major species (10 % or more of sale value; one stumpage rate, so value share equals volume share) and 20 plots minimum. Measured trees are tally trees with a valid code, DBH in range and a height. The basal-area error is reference only. Stand-summary BA counts every tallied tree (plot count x BAF); the stand and stock tables drop records flagged by QA, such as an out-of-range DBH, so the two can differ by a few tenths. Sources and URLs: cruise_data.xlsx, Standards and assumptions.

TABLE 2. STAND TABLE BY 4-INCH DBH CLASS, PER ACRE

DBH class, in	Trees	TPA	BA	cu ft	MBF
8-11.9	1	1.6	1.2	24	0.1
12-15.9	12	12.2	14.1	358	1.8
16-19.9	17	11.5	20.0	577	2.9
20-23.9	41	18.1	48.2	1464	7.3
24-27.9	29	9.6	34.1	1113	5.6
28-31.9	34	8.2	40.0	1409	7.1
32-35.9	21	4.0	24.7	907	4.6
36-39.9	19	2.9	22.4	873	4.4
40-43.9	13	1.6	15.3	615	3.1
44-47.9	7	0.7	8.2	380	1.9
48-51.9	5	0.4	5.9	282	1.4
52-55.9	1	0.1	1.2	65	0.3
56-59.9	3	0.2	3.5	180	0.9
60-63.9	2	0.1	2.4	121	0.6
All	205	71.2	241.2	8366	42.0

TABLE 3. STOCK TABLE BY SPECIES, PER ACRE

Species	Trees	TPA	BA	QMD	cu ft	MBF
WF	70	21.1	82.4	26.7	3044	15.3
PP	51	21.5	60.0	22.6	2164	10.9
IC	36	12.1	42.4	25.4	1246	6.3
DF	29	10.2	34.1	24.8	1107	5.6
SP	19	6.3	22.4	25.5	806	4.0
All	205	71.2	241.2	24.9	8366	42.0

Trees: tally trees. TPA, BA (sq ft/ac), cubic feet and Scribner MBF per acre by BAF expansion. Species codes: PP ponderosa pine, WF white fir, DF Douglas-fir, SP sugar pine, IC incense-cedar.

Unit 406 plots, 300 ft grid (orange: QA flag)

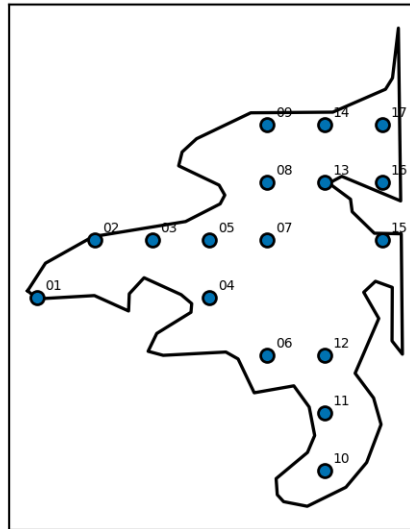


Figure 1. Plot locations with QA status.

QA findings: none; unit accepted; 3 more plots to reach the R5 20-plot minimum (sec. 41.3)

TABLE 4. PLOT LIST

Plot	Trees in	BA sq ft/ac	Slope %	Status
406-01	13	260	39	ok
406-02	12	240	45	ok
406-03	12	240	66	ok
406-04	13	260	40	ok
406-05	11	220	59	ok
406-06	14	280	35	ok
406-07	12	240	54	ok
406-08	11	220	67	ok
406-09	10	200	53	ok
406-10	11	220	61	ok
406-11	12	240	45	ok
406-12	11	220	35	ok
406-13	12	240	37	ok
406-14	13	260	48	ok
406-15	11	220	44	ok
406-16	13	260	41	ok
406-17	14	280	44	ok